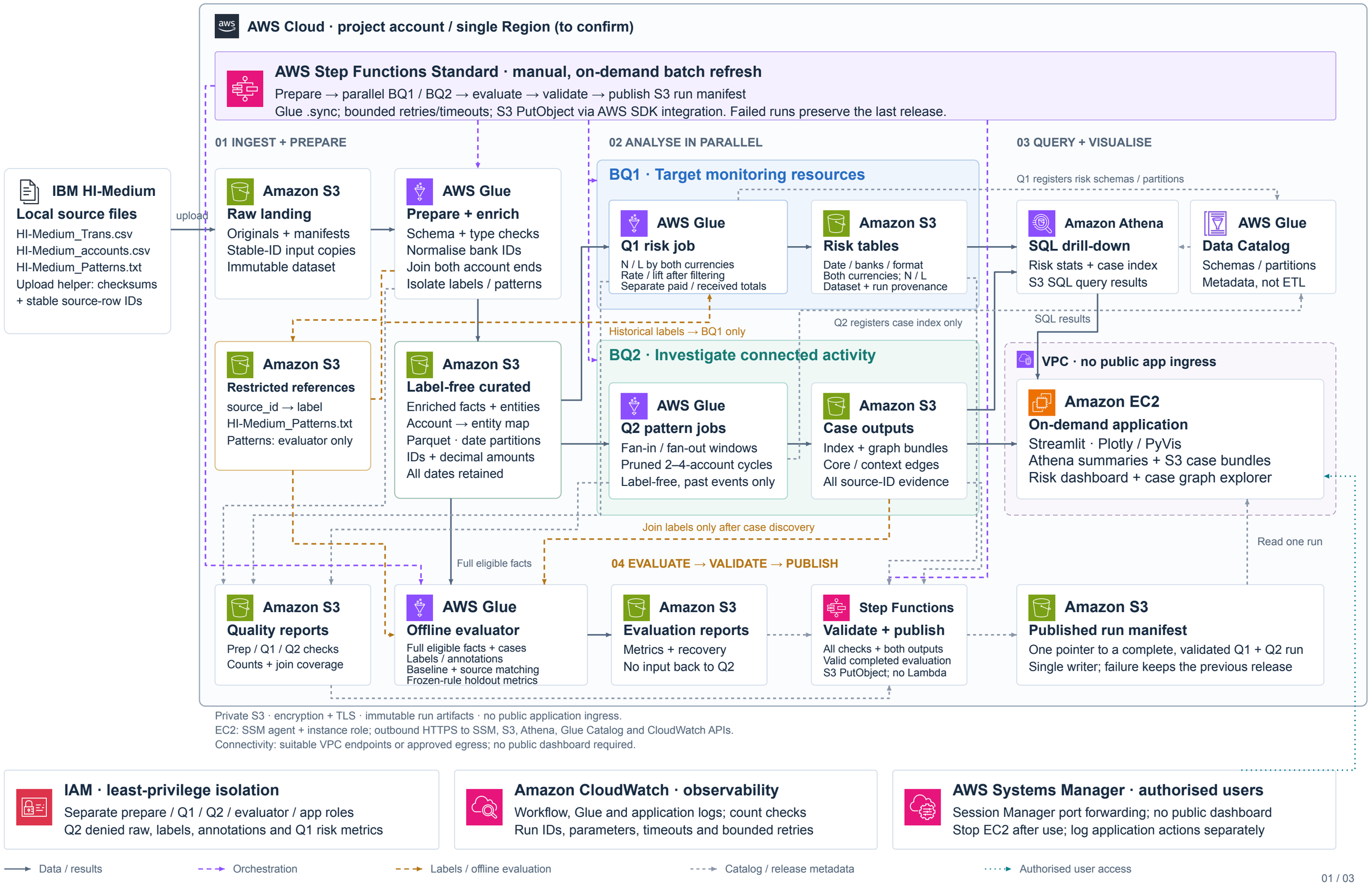
Proposed AWS architecture

PROPOSED · NOT DEPLOYED

AML monitoring allocation + connected-account investigation

IBM HI-Medium · 31,898,238 transactions · synthetic data



Private S3 · encryption + TLS · immutable run artifacts · no public application ingress.
EC2: SSM agent + instance role; outbound HTTPS to SSM, S3, Athena, Glue Catalog and CloudWatch APIs.
Connectivity: suitable VPC endpoints or approved egress; no public dashboard required.

IAM · least-privilege isolation
Separate prepare / Q1 / Q2 / evaluator / app roles
Q2 denied raw, labels, annotations and Q1 risk metrics

Amazon CloudWatch · observability
Workflow, Glue and application logs; count checks
Run IDs, parameters, timeouts and bounded retries

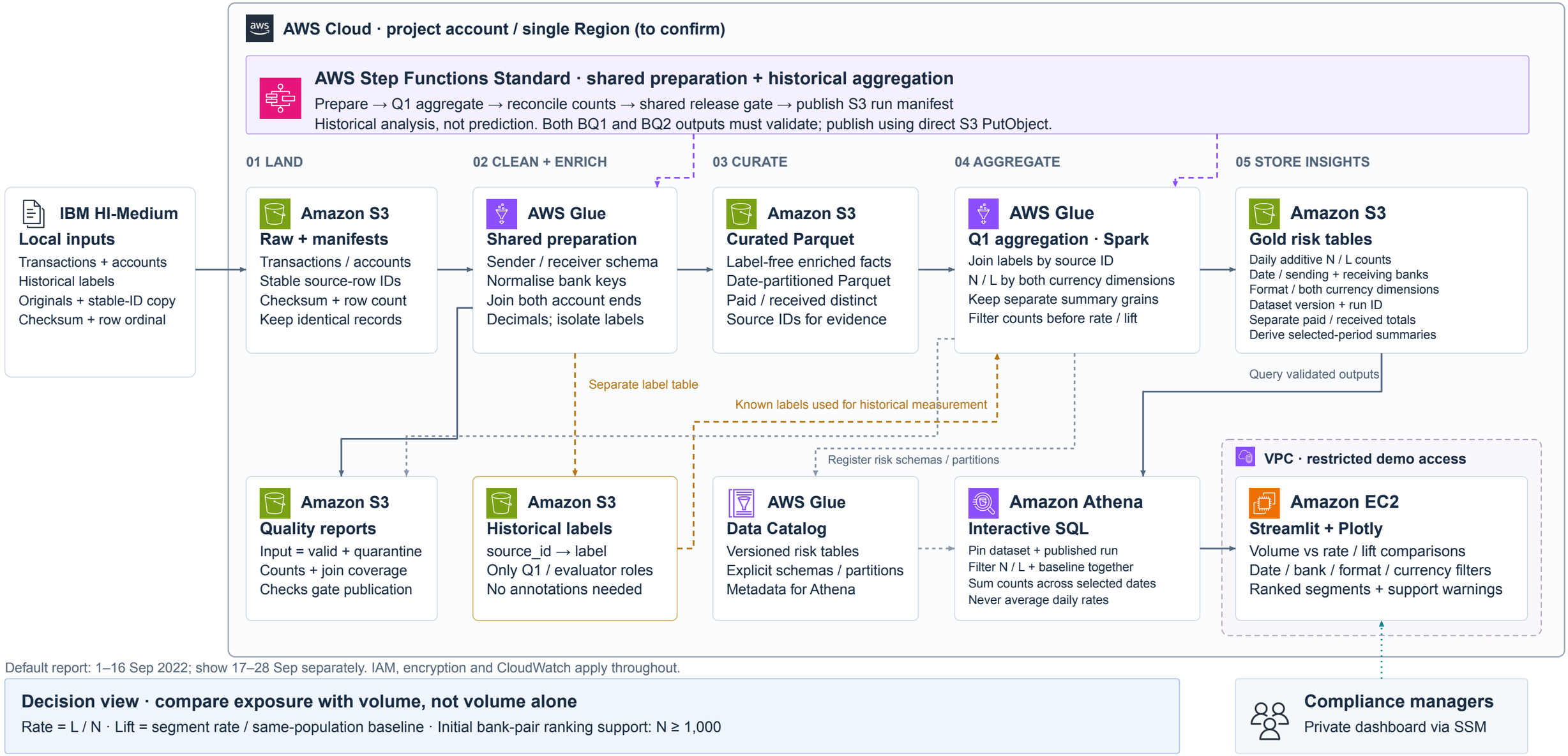
AWS Systems Manager · authorised users
Session Manager port forwarding; no public dashboard
Stop EC2 after use; log application actions separately

Data / results · Orchestration · Labels / offline evaluation · Catalog / release metadata · Authorised user access

BQ1 · Monitoring allocation

PROPOSED · NOT DEPLOYED

Which payment formats and bank-to-bank segments have disproportionately high historical laundering exposure, after accounting for transaction volume, so compliance managers can target monitoring resources?



Default report: 1–16 Sep 2022; show 17–28 Sep separately. IAM, encryption and CloudWatch apply throughout.

Decision view · compare exposure with volume, not volume alone

Rate = L / N · Lift = segment rate / same-population baseline · Initial bank-pair ranking support: N ≥ 1,000

Compliance managers

Private dashboard via SSM

Data / results · Orchestration · Labels / offline evaluation · Catalog / release metadata · Authorised user access

BQ2 · Connected-account investigation

PROPOSED · NOT DEPLOYED

Which connected groups of accounts and entities exhibit coordinated transfer patterns, such as fan-in, fan-out or circular transfers, that warrant investigation beyond an individual transaction?

IBM HI-Medium
Local inputs
HI-Medium_Trans.csv
HI-Medium_accounts.csv
HI-Medium_Patterns.txt
Checksum + row ordinal

AWS Cloud · project account / single Region (to confirm)

AWS Step Functions Standard · discover first, evaluate afterwards
Prepare → daily snapshots + pattern tasks → case evidence → isolated evaluation → validated S3 publication
Label-free detector; frozen rules before holdout; only events at or before each snapshot cutoff enter its features.

01 SHARED INGESTION

Amazon S3
Raw landing
Originals + manifests
Stable-ID input copies
Trusted prep role only

AWS Glue
Prepare + enrich
Schema + key checks
Join accounts / entities
Isolate labels / patterns

S3 · preparation / Q2 quality reports
Account for every input row
Keep unknown entities; no duplicate joins
Version source IDs, code and parameters

02 LABEL-FREE GRAPH INPUTS

Amazon S3
Label-free curated
Date-partitioned Parquet facts
Account → entity mappings
No Is Laundering column
Full eligible facts → evaluator

Directed account graph
Node = bank + account
Edge = source transaction
Entity = owner grouping
Keep parallel edges + IDs

03 DETECT MOTIFS

AWS Glue · Spark
Q2 pattern jobs

Fan-in / fan-out
≥5 distinct non-self counterparties
24-hour / 7-day rolling windows
Spark windowed aggregations

Bounded directed cycles
2–4 distinct accounts; 7-day window
Pruned edge joins; log exclusions
Self-loops are not cycle candidates

Label-free · past events only

04 INVESTIGATE
Q2 registers case-index schemas / partitions

Amazon S3
Gold case store
Case index + per-case graph JSON
Rule / cutoff / participants / entity links
Core vs context edges; full source-ID evidence

S3 bundles → explorer

Amazon Athena
Case-index SQL
Date / rule filters
Published tables

AWS Glue
Data Catalog
Index schema
Run partitions

VPC · no public app ingress

Amazon EC2
Streamlit + PyVis
Case graph + triggered-rule explanation
Visible cap: 100 nodes / 500 edges
Export full evidence beyond display cap

OFFLINE EVALUATION · no feedback to Q2

Amazon S3
Evaluation references
HI-Medium_Patterns.txt
+ transaction labels
Match annotated attempts

AWS Glue
Offline evaluator
Full eligible facts + cases
Labels / pattern annotations
Amount baseline + source matching
Frozen-rule holdout

Amazon S3
Evaluation reports
Precision / recall + recovery
Pruning + evidence coverage
For release gate, not Q2 rules

IAM access
Q2: curated facts
+ account maps only.
Deny raw / labels /
patterns / Q1 scores.

Step Functions publishes validated shared-run metadata directly to S3. IAM, encryption and CloudWatch apply throughout.

Evaluation protocol · patterns are investigation leads, not proof of laundering
1–8 Sep: develop / calibrate → 9–16 Sep: frozen-rule holdout → 17–28 Sep: separate sparse-tail study
Retain full evidence; a structural cycle is not automatically a time-ordered path or the same funds returning.

AML investigators
Private case explorer via SSM
Inspect links + export evidence

Data / results · Orchestration · Labels / offline evaluation · Catalog / release metadata · Authorised user access